

FITBUDDY

AI-Powered Personalized Workout and Wellness Assistant

PHASE 04

Project Planning Phase

Organize work into milestones, assign implementation responsibilities, identify dependencies, and define acceptance criteria.

Prepared for phase-wise submission

30 September 2026

Phase objective: Organize work into milestones, assign implementation responsibilities, identify dependencies, and define acceptance criteria.

1. Work Breakdown Structure

- WBS-1: Confirm problem statement and scope.
- WBS-2: Define requirements and acceptance criteria.
- WBS-3: Design architecture, database, APIs, and AI schemas.
- WBS-4: Implement backend and persistence.
- WBS-5: Implement frontend templates and styling.
- WBS-6: Integrate AI and mock mode.
- WBS-7: Create automated tests and perform verification.
- WBS-8: Prepare documentation and demonstration assets.

2. Suggested Milestones

- M1 - Concept approved: problem, users, scope defined.
- M2 - Requirements approved: FR/NFR and inputs/outputs documented.
- M3 - Design approved: architecture and schemas ready.
- M4 - Core development complete: routes, models, templates, services connected.
- M5 - AI/mock flow verified: plan generation and revision work.
- M6 - Test and documentation complete: automated tests and setup guide ready.
- M7 - Demo ready: recording sequence and final outputs prepared.

3. Dependencies

- Python environment and virtual environment.
- Packages from requirements.txt.
- Writable data directory for SQLite.
- Gemini API key only for live AI mode; mock mode removes this dependency for testing.
- VS Code or another Python-capable IDE for development and debugging.

4. Risk Register

- AI model availability risk: model identifiers or account access may vary; use environment variables and mock mode.
- API credential risk: store secrets in .env and never commit the file.
- Data/security risk: the local users page has no authentication in the current implementation.
- Prompt/response risk: enforce structured schemas and review generated wellness guidance.
- Environment risk: Python path, PowerShell execution policy, or missing dependencies can block setup.

5. Acceptance Criteria

- Application starts through Uvicorn without import/configuration errors.
- Home page accepts valid profile data.
- Workout generation returns exactly seven days.
- Nutrition tip is displayed.
- Feedback produces an updated plan.
- User record is visible in the users page.
- Automated tests pass in mock mode.

Suggested Planning Schedule

Work period	Focus
Week 1	Ideation + requirements
Week 2	Architecture + database + UI design
Week 3	Backend + templates + persistence
Week 4	AI integration + feedback revision
Week 5	Testing + bug fixing
Week 6	Documentation + demo preparation

Phase Deliverables / Evidence

- Work breakdown structure
- Milestones and dependencies
- Risk register and acceptance criteria

Submission Note

This phase document is intended to be submitted together with the corresponding FitBuddy source project and evidence captured during execution. Keep API keys, personal credentials, and other secrets out of screenshots and submissions.